Real Analysis and Measure Theory

McGill MATH 564, Part II

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Part II of the course notes: the lectures after the midterm, written up from Professor Toth's handwritten lecture notes. Section numbers continue those of Part I, which has Sections 1–12: Section n covers Lecture n, including its second part. Slips in the original notes have been corrected, routine steps left as exercises are filled in, and facts the lectures use without proof are proved where first needed. Paragraphs labelled Explanation. are additions and are not from the lectures.

Definitions **Results** **Examples**

remarks, notes and explanations run inline, with an underlined label

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13 Applications of the Monotone Convergence Theorem

Throughout, $(X, \mathcal{M}, \mu)$ is a measure space, $L^+ = \{f : X \to [0,\infty] \mid f \in \mathrm{Meas}(X)\}$, and $\int f$ means $\int_X f\, d\mu$. This lecture applies the MCT and its series version, MCT + Summation (Part I, Section 12).

13.1 Null Sets and Integrals

Sets of measure zero have no bearing on integrals. The precise version is the following.

Proposition (Zero integral)

Let $f \in L^+$. Then $\int f = 0$ if and only if $f = 0$ a.e.

Proof

Simple functions. Let $\varphi \in \mathrm{Sim}(X) \cap L^+$, written $\varphi = \sum_{j=1}^{n} a_j \chi_{E_j}$ with $a_j \geq 0$ and $E_j \in \mathcal{M}$ disjoint, $\bigcup_j E_j = X$. Then $\int \varphi = \sum_j a_j\, \mu(E_j)$ is a sum of non-negative terms, so $\int \varphi = 0$ iff $\mu(E_j) = 0$ for every j with $a_j > 0$. Since $\{\varphi \neq 0\} = \bigcup_{a_j>0} E_j$, this holds iff $\mu(\{\varphi \neq 0\}) = 0$, i.e. iff $\varphi = 0$ a.e.

($\Leftarrow$) Let $f = 0$ a.e. If $\varphi \in \mathrm{Sim}(X)$ and $0 \leq \varphi \leq f$, then $\{\varphi \neq 0\} \subseteq \{f \neq 0\}$, so $\varphi = 0$ a.e., and $\int \varphi = 0$ by the simple case. Taking the supremum over all such φ gives $\int f = 0$.

($\Rightarrow$) Let $\int f = 0$, and set $E_n := \{x \mid f(x) > \frac{1}{n}\} \in \mathcal{M}$, so that $\{f > 0\} = \bigcup_{n=1}^{\infty} E_n$. Since $f > \frac{1}{n}$ on E_n and $f \geq 0$ elsewhere, $f \geq \frac{1}{n}\chi_{E_n}$, and by monotonicity

$$0 = \int f \geq \frac{1}{n} \int \chi_{E_n} = \frac{1}{n}\, \mu(E_n).$$

Hence $\mu(E_n) = 0$ for all n, and by subadditivity $\mu(\{f > 0\}) \leq \sum_n \mu(E_n) = 0$, i.e. $f = 0$ a.e.

So the MCT is insensitive to sets of measure zero:

Theorem (MCT, a.e. version)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^+$ and $f \in L^+$ with $f_n(x) \nearrow f(x)$ for a.e. $x \in X$. That is, there is a null set N such that for every $x \notin N$, $f_n(x) \leq f_{n+1}(x)$ for all n and $f_n(x) \to f(x)$. Then

$$\int f = \lim_{n\to\infty} \int f_n.$$

Proof

Let $E := X \setminus N$, so $\mu(E^c) = 0$. First, $\int h = \int h\chi_E$ for every $h \in L^+$. Indeed, $h = h\chi_E + h\chi_{E^c}$ with both terms in L^+, and $h\chi_{E^c} = 0$ outside the null set E^c, so $\int h\chi_{E^c} = 0$ by the Proposition (Zero integral). By additivity of the integral on L^+ (MCT + Summation, Part I, Section 12),

$$\int h = \int h\chi_E + \int h\chi_{E^c} = \int h\chi_E.$$

(The lecture wrote this step as $0 = \int (f - f\chi_E) = \int f - \int f\chi_E$, which needs $\int f < \infty$; writing h as a sum avoids the subtraction.)

Now $f_n\chi_E \nearrow f\chi_E$ at every point of X: on E by assumption, and on E^c all these functions vanish. By the MCT (Part I, Section 12) and the first step for $h = f$ and $h = f_n$,

$$\int f = \int f\chi_E = \lim_{n\to\infty} \int f_n\chi_E = \lim_{n\to\infty} \int f_n.$$

Corollary (Null sets do not matter)

Let $f, g \in L^+$ with $f = g$ a.e. Then $\int f = \int g$.

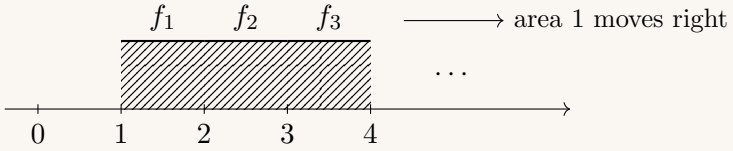
Proof

Apply the MCT, a.e. version, with $f_n := g$ for all n. (The lecture uses this in the form "sets of measure zero do not matter".)

Remark. The MCT assumes $f_n \in L^+$ and $f_n \nearrow f$. If monotonicity is dropped, the conclusion can fail. Counterexamples are easy to construct by letting the area under the graphs escape to infinity.

Counterexample (Escaping mass)

On $(\mathbb{R}, \mathcal{L}, m)$, let $f_n := \chi_{[n,n+1)}$.



Then $f_n \to 0$ pointwise: for fixed $x \in \mathbb{R}$, $x \notin [n, n+1)$ once $n > x$. But $\int f_n\, dm = m([n, n+1)) = 1$ for every n, so

$$\int_{\mathbb{R}} \lim_{n\to\infty} f_n\, dm = 0 \neq 1 = \lim_{n\to\infty} \int_{\mathbb{R}} f_n\, dm.$$

The sequence $\{f_n\}$ is not increasing, so the MCT does not apply.

13.2 Fatou's Lemma

Question. Given $\{f_n\}_{n=1}^{\infty} \subseteq L^+$, what can be said about general (not monotone) limits?

Theorem (Fatou's lemma)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^+$ be any sequence; monotonicity is not required. Then

$$\int \liminf_{n\to\infty} f_n\, d\mu \leq \liminf_{n\to\infty} \int f_n\, d\mu.$$

Proof

The idea is to apply the MCT to $g_k := \inf_{n\geq k} f_n$. Each g_k lies in L^+ (Part I, Section 10), $g_k \leq g_{k+1}$

since the infimum is taken over fewer terms, and $g_k \nearrow \liminf_n f_n$ by definition of $\liminf$ (Part I, Section 10). So by the MCT,

$$\lim_{k\to\infty} \int g_k = \int \liminf_{n\to\infty} f_n. \tag{1}$$

For every k and every $j \geq k$, $g_k \leq f_j$ by definition of the infimum, so $\int g_k \leq \int f_j$ by monotonicity. Hence

$$\int g_k \leq \inf_{j\geq k} \int f_j. \tag{2}$$

As $k \to \infty$, the right side of (2) tends to $\liminf_j \int f_j$, so (1) and (2) give the claim.

Remark. The lecture mentions a "sup" version of Fatou's lemma, to be discussed later.

Explanation. The inequality can be strict. For $f_n = \chi_{[n,n+1)}$ in the Counterexample (Escaping mass) above, $\int \liminf_n f_n = 0 < 1 = \liminf_n \int f_n$. Mass can disappear in the limit, but no mass can appear.

Corollary (Fatou, a.e. version)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^+$ and $f \in L^+$ with $f_n \to f$ a.e. Then

$$\int f \leq \liminf_{n\to\infty} \int f_n, \qquad \text{i.e.} \qquad \int \lim_{n\to\infty} f_n \leq \liminf_{n\to\infty} \int f_n.$$

Proof

Wherever $f_n \to f$, $\liminf_n f_n = \lim_n f_n = f$. So $f = \liminf_n f_n$ a.e., and both functions lie in L^+. By the Corollary (Null sets do not matter) and Fatou's lemma, $\int f = \int \liminf_n f_n \leq \liminf_n \int f_n$.

13.3 Functions with Finite Integral

Proposition (Finite integral)

Let $f \in L^+$ with $\int f < \infty$. Then:

① $\mu(\{x \mid f(x) = \infty\}) = 0$, i.e. $f < \infty$ a.e.;
② $\{x \mid f(x) > 0\}$ is σ-finite: it is a countable union of sets in $\mathcal{M}$ of finite measure.

Proof

① Let $F_\infty := \{f = \infty\} \in \mathcal{M}$. For every $M > 0$ we have $f \geq M\chi_{F_\infty}$, so by monotonicity $M\mu(F_\infty) \leq \int f < \infty$. This holds for every $M > 0$, which forces $\mu(F_\infty) = 0$.

② Let $E_n := \{f > \frac{1}{n}\} \in \mathcal{M}$. Then $\{f > 0\} = \bigcup_{n=1}^{\infty} E_n$, and $f \geq \frac{1}{n}\chi_{E_n}$ gives $\mu(E_n) \leq n \int f < \infty$ for every n.

Explanation. The proofs of the Propositions (Zero integral) and (Finite integral) use one estimate. For $f \in L^+$ and $a > 0$, $f \geq a\chi_{\{f>a\}}$, so

$$\mu(\{f > a\}) \leq \frac{1}{a} \int f$$

(*Chebyshev's inequality*). It turns a bound on the integral into a bound on the size of the set where f is large.

13.4 Integrable Functions

Now let $f : X \to \mathbb{R}$ with $f \in \mathrm{Meas}(X)$; sign changes are allowed. Recall that $f = f^+ - f^-$, where the positive and negative parts $f^\pm$ lie in L^+ (Part I, Section 11).

Definition

A function $f : X \to \mathbb{R}$ in $\mathrm{Meas}(X)$ is *integrable*, written $f \in L^1(X, \mathcal{M}, \mu)$ (or $L^1(\mu)$, or L^1), if $\int f^+ \, d\mu < \infty$ and $\int f^- \, d\mu < \infty$. Its *integral* is then

$$\int_X f \, d\mu := \int_X f^+ \, d\mu - \int_X f^- \, d\mu.$$

A function $f : X \to \mathbb{C}$ in $\mathrm{Meas}(X)$ is integrable if $\mathrm{Re}\, f$ and $\mathrm{Im}\, f$ are, and then

$$\int_X f \, d\mu := \int_X \mathrm{Re}\, f \, d\mu + i \int_X \mathrm{Im}\, f \, d\mu.$$

As before, $\int_E f \, d\mu := \int_X f \chi_E \, d\mu$ for $E \in \mathcal{M}$.

Note. Since $|f| = f^+ + f^-$ (Part I, Section 11) and the integral is additive on L^+, $\int |f| = \int f^+ + \int f^-$. Hence

$$L^1(X, \mathcal{M}, \mu) = \Big\{ f \in \mathrm{Meas}(X) \;\Big|\; \int_X |f| \, d\mu < \infty \Big\}.$$

This also holds for complex f, since $|\mathrm{Re}\, f|, |\mathrm{Im}\, f| \le |f| \le |\mathrm{Re}\, f| + |\mathrm{Im}\, f|$.

Corollary (Null sets, integrable case)

Let $f \in L^1$ and $g \in \mathrm{Meas}(X)$ with $f = g$ a.e. Then $g \in L^1$ and $\int g = \int f$.

> ***Proof***
>
> The lectures use this without stating it. For real f, $g^\pm = f^\pm$ a.e., so the Corollary (Null sets do not matter) applied to f^+ and to f^- gives $\int g^\pm = \int f^\pm < \infty$. For complex f, apply the real case to the real and imaginary parts.

Remark. To prove results about the integral of a general $f \in \mathrm{Meas}(X)$, where sign changes are allowed, one needs $\int f^\pm < \infty$, i.e. $f \in L^1$. The notes of Lecture 13 treat real f only; the recap at the start of Lecture 14 mentions complex functions, and the definition via real and imaginary parts is added here.

Next. Dominated convergence.

14 The Dominated Convergence Theorem

14.1 Linearity and the Basic Estimate

Lecture 14 starts by recalling Fatou's lemma and the definition of L^1 (Section 13). The basic estimate and the proof of the DCT below use that the integral is linear on L^1. The lecture takes this for granted, so it is proved first.

Proposition (Linearity)

Let $f, g \in L^1(X, \mathcal{M}, \mu)$ and $a, b \in \mathbb{C}$. Then $af + bg \in L^1$ and

$$\int (af + bg) = a \int f + b \int g.$$

In particular, the real-valued and the complex-valued functions in L^1 each form a vector space.

Proof

First, $af + bg \in \mathrm{Meas}(X)$ (Part I, Section 10). For $c \geq 0$ and $h \in L^+$ we have $\int ch = c \int h$: take simple $\varphi_n \nearrow h$ (approximation theorem, Part I, Section 11); then $c\varphi_n \nearrow ch$, and the MCT together with $\int c\varphi_n = c \int \varphi_n$ (Part I, Section 11) gives $\int ch = \lim_n c \int \varphi_n = c \int h$. Since $|af + bg| \leq |a||f| + |b||g|$, monotonicity and additivity on L^+ give

$$\int |af + bg| \leq |a| \int |f| + |b| \int |g| < \infty,$$

so $af + bg \in L^1$. It remains to prove additivity and homogeneity, first for real f, g and a.

Sums. Let $h := f + g$. Then $h^+ - h^- = f^+ - f^- + g^+ - g^-$, so

$$h^+ + f^- + g^- = h^- + f^+ + g^+,$$

an identity between functions in L^+ with finite values. Integrating both sides, by additivity on L^+,

$$\int h^+ + \int f^- + \int g^- = \int h^- + \int f^+ + \int g^+.$$

All six integrals are finite, so we may rearrange to get $\int h = \int f + \int g$.

Scalars. If $a \geq 0$, then $(af)^\pm = af^\pm$, so $\int af = a \int f^+ - a \int f^- = a \int f$. If $a < 0$, then $(af)^+ = |a| f^-$ and $(af)^- = |a| f^+$, so $\int af = |a| \int f^- - |a| \int f^+ = a \int f$.

Complex case. Since $\mathrm{Re}(f + g) = \mathrm{Re}\, f + \mathrm{Re}\, g$, and likewise for Im, the real case gives $\int (f + g) = \int f + \int g$. For $a = s + it$ with $s, t \in \mathbb{R}$, $\mathrm{Re}(af) = s\,\mathrm{Re}\, f - t\,\mathrm{Im}\, f$ and $\mathrm{Im}(af) = t\,\mathrm{Re}\, f + s\,\mathrm{Im}\, f$, so by the real case

$$\int af = \Big(s \int \mathrm{Re}\, f - t \int \mathrm{Im}\, f\Big) + i\Big(t \int \mathrm{Re}\, f + s \int \mathrm{Im}\, f\Big) = (s + it) \int f.$$

Proposition (Basic estimate)

Let $f \in L^1(X, \mathcal{M}, \mu)$. Then

$$\left| \int f \, d\mu \right| \leq \int |f| \, d\mu.$$

Proof

The lecture states this without proof. For real f, using $|f| = f^+ + f^-$ and additivity on L^+,

$$\left|\int f\right| = \left|\int f^+ - \int f^-\right| \le \int f^+ + \int f^- = \int |f|.$$

For complex f we may assume $\int f \neq 0$. Let $\alpha := \overline{\operatorname{sgn}\left(\int f\right)}$ (Part I, Section 11), so $|\alpha| = 1$ and $\alpha \int f = |\int f|$. By linearity $|\int f| = \int \alpha f$. This number is real, so it equals its real part, $\int \operatorname{Re}(\alpha f)$. Finally $\operatorname{Re}(\alpha f) \le |\alpha f| = |f|$, and for real functions in L^1 the integral is monotone, by linearity: $\int |f| - \int \operatorname{Re}(\alpha f) = \int \big(|f| - \operatorname{Re}(\alpha f)\big) \ge 0$. Hence $|\int f| \le \int |f|$.

14.2 The Dominated Convergence Theorem

A very basic and important result is the following.

Theorem (DCT)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^1(X, \mathcal{M}, \mu)$ be real- or complex-valued, and suppose that:

(i) $f_n \to f$ a.e., for some function f on X;
(ii) there is $g \in L^1$ with $|f_n| \le g$ a.e. for all $n \ge 1$ (so g is independent of n).

Then $f \in L^1$ and

$$\int f \, d\mu = \lim_{n\to\infty} \int f_n \, d\mu.$$

Proof

Reduction. $f \in \operatorname{Meas}(X)$, as an a.e. limit of measurable functions, since μ is complete (Part I, Sections 6 and 11). The set where (i) fails, or where $|f_n| > g$ for some n, lies in a countable union of null sets, hence in a null set $N \in \mathcal{M}$. Replacing f_n, f, g by $f_n\chi_{N^c}$, $f\chi_{N^c}$, $g\chi_{N^c}$ changes neither membership in L^1 nor any integral (Corollary (Null sets, integrable case) in Section 13). So we may assume that (i) and (ii) hold everywhere; in particular $g \ge 0$. Then $|f| = \lim_n |f_n| \le g$, so $\int |f| \le \int g < \infty$ and $f \in L^1$.

Real case. The idea is to apply Fatou's lemma to two different sequences in L^+. Since $|f_n| \le g$, we have $f_n \le g$ and $-f_n \le g$, i.e.

$$g - f_n \ge 0 \qquad \text{and} \qquad g + f_n \ge 0 \qquad \text{for all } n.$$

So $\{g - f_n\}$ and $\{g + f_n\}$ lie in L^+, and also in L^1 by linearity. Since $\lim_n f_n = f$, we have $\liminf_n (g + f_n) = g + f$. By linearity, Fatou's lemma and linearity again,

$$\begin{aligned}\int g + \int f = \int (g + f) &= \int \liminf_{n\to\infty} (g + f_n) \\ &\le \liminf_{n\to\infty} \int (g + f_n) = \int g + \liminf_{n\to\infty} \int f_n.\end{aligned}$$

Since $\int g < \infty$, we can subtract it:

$$\int f \le \liminf_{n\to\infty} \int f_n. \tag{$\dagger$}$$

Now run the same argument with $\{g-f_n\}$. For a real sequence (a_n), $\liminf_n(-a_n) = -\limsup_n a_n$, since $\inf_{n\geq k}(-a_n) = -\sup_{n\geq k} a_n$. So

$$\int g - \int f = \int \liminf_{n\to\infty}(g-f_n) \leq \liminf_{n\to\infty} \int (g-f_n) = \int g - \limsup_{n\to\infty} \int f_n,$$

and subtracting $\int g$ again gives

$$\limsup_{n\to\infty} \int f_n \leq \int f. \tag{‡}$$

By (†) and (‡),

$$\limsup_{n\to\infty} \int f_n \leq \int f \leq \liminf_{n\to\infty} \int f_n \leq \limsup_{n\to\infty} \int f_n,$$

so all these are equal: $\lim_n \int f_n$ exists and equals $\int f$.

Complex case. (The lecture proves the real case only.) We have $\operatorname{Re} f_n \to \operatorname{Re} f$ and $|\operatorname{Re} f_n| \leq |f_n| \leq g$, and likewise for Im. So by the real case,

$$\int f = \int \operatorname{Re} f + i \int \operatorname{Im} f = \lim_{n\to\infty} \int \operatorname{Re} f_n + i \lim_{n\to\infty} \int \operatorname{Im} f_n = \lim_{n\to\infty} \int f_n.$$

Explanation. The dominating function rules out the escaping mass of the Counterexample in Section 13. If $g \geq \chi_{[n,n+1)}$ for all n, then $g \geq \chi_{[1,\infty)}$, so $\int g = \infty$: no integrable g dominates that sequence.

14.3 The Series Version of the DCT

Theorem (Series version of the DCT)

Let $\{f_n\}_{n=1}^{\infty} \subseteq L^1(X, \mathcal{M}, \mu)$ with

$$\sum_{n=1}^{\infty} \int |f_n|\, d\mu < \infty.$$

Then $\sum_{n=1}^{\infty} f_n(x)$ converges for a.e. $x \in X$, its sum f (defined a.e.) lies in L^1, and

$$\int \sum_{n=1}^{\infty} f_n\, d\mu = \sum_{n=1}^{\infty} \int f_n\, d\mu.$$

Proof

Let $G := \sum_{n=1}^{\infty} |f_n| \in L^+$. By MCT + Summation (Part I, Section 12) we can interchange summation and integration for the $|f_n|$:

$$\int G = \sum_{n=1}^{\infty} \int |f_n| < \infty.$$

By ① of the Proposition (Finite integral) in Section 13, $G < \infty$ a.e.: the set $E := \{G < \infty\} \in \mathcal{M}$ has $\mu(E^c) = 0$. For $x \in E$ the series $\sum_n f_n(x)$ converges absolutely, hence converges. Set $f(x) := \sum_n f_n(x)$ for $x \in E$ and $f(x) := 0$ for $x \notin E$; by the Corollary (Null sets, integrable case)

in Section 13, other choices on E^c change nothing below.

Let $S_N := \sum_{n=1}^{N} f_n$ be the partial sums. Then $S_N\chi_E \to f$ everywhere, so $f \in \mathrm{Meas}(X)$ (Part I, Section 10). Moreover $G\chi_E \in L^1$, since $\int G\chi_E \le \int G < \infty$, and on E

$$|S_N| = \Big|\sum_{n=1}^{N} f_n\Big| \le \sum_{n=1}^{N} |f_n| \le G = G\chi_E \qquad \text{for all } N \ge 1.$$

So $|S_N| \le G\chi_E$ a.e. and $S_N \to f$ a.e. By the DCT, $f \in L^1$ and, using linearity for the finite sums,

$$\int f = \lim_{N\to\infty} \int S_N = \lim_{N\to\infty} \sum_{n=1}^{N} \int f_n = \sum_{n=1}^{\infty} \int f_n.$$

Explanation. Compared with MCT + Summation (Part I, Section 12), the f_n may change sign or be complex. The price is the hypothesis $\sum_n \int |f_n| < \infty$, which is exactly what makes $G = \sum_n |f_n|$ an integrable dominating function for the partial sums.

14.4 Density Results

Idea. When proving results about L^1, it is very useful in practice to replace $f \in L^1$ by a "nice" approximation.

Notation. For $f \in L^1(X, \mathcal{M}, \mu)$, write $\|f\|_{L^1} := \int_X |f|\, d\mu$.

Explanation. From $|f + g| \le |f| + |g|$ and the properties of the integral on L^+, $\|f + g\|_{L^1} \le \|f\|_{L^1} + \|g\|_{L^1}$ and $\|af\|_{L^1} = |a|\|f\|_{L^1}$. But $\|f\|_{L^1} = 0$ only says that $f = 0$ a.e. (Proposition (Zero integral) in Section 13, applied to $|f|$), so $\|\cdot\|_{L^1}$ is a norm only once functions that agree a.e. are identified.

Theorem (Density)

Let $(X, \mathcal{M}, \mu)$ be a measure space, $f \in L^1(X, \mathcal{M}, \mu)$ and $\varepsilon > 0$.

- (a) There is a simple function $\varphi = \sum_{j=1}^{N} a_j \chi_{E_j} \in \mathrm{Sim}(X)$ with $\varphi \in L^1$ and $\|f - \varphi\|_{L^1} < \varepsilon$. In other words, $\mathrm{Sim}(X) \cap L^1$ is dense in L^1 for $\|\cdot\|_{L^1}$.
- (b) If $\mu = \mu_F$ is a Lebesgue–Stieltjes measure on $\mathbb{R}$ (Part I, Section 8), the sets E_j in (a) can be taken to be finite unions of open intervals. Moreover, there is $g \in C_c^0(\mathbb{R})$, i.e. g is continuous and $g = 0$ outside some bounded interval, with $\|f - g\|_{L^1} < \varepsilon$.

Proof (a)

The candidates are the simple functions that approximate f pointwise. By the approximation theorem (Part I, Section 11) there are $\varphi_n \in \mathrm{Sim}(X)$ with

(i) $0 \le |\varphi_1| \le |\varphi_2| \le \cdots \le |f|$;
(ii) $\varphi_n \to f$ pointwise.

By (i), each $\varphi_n \in L^1$. By the triangle inequality and (i),

$$|\varphi_n - f| \le |\varphi_n| + |f| \le 2|f| \in L^1 \qquad \text{for all } n,$$

and $|\varphi_n - f| \to 0$ pointwise by (ii). So the DCT applies to the sequence $\{|\varphi_n - f|\}$:

$$\lim_{n\to\infty} \int |\varphi_n - f| = \int \lim_{n\to\infty} |\varphi_n - f| = 0.$$

Hence there is $n(\varepsilon)$ with $\|f - \varphi_n\|_{L^1} < \varepsilon$ for all $n \ge n(\varepsilon)$.

Proof (b)

The lecture left Steps 2–4 below as exercises, sketching Step 3 for bounded intervals. Let $\mu = \mu_F$ on $\mathcal{M}_\mu$, and fix $\varepsilon > 0$. We split ε into three parts.

Step 1: the sets E_j have finite measure. Take $\varphi = \varphi_n$ from the proof of ⓐ with $\|f - \varphi\|_{L^1} < \varepsilon/3$, and write $\varphi = \sum_{j=1}^N a_j \chi_{E_j}$ with the E_j disjoint and, after dropping zero terms, all $a_j \neq 0$. Integrating over E_k, where $|\varphi| = |a_k|$, and using $|\varphi| \le |f|$,

$$|a_k|\,\mu(E_k) = \int_{E_k} |\varphi| \le \int_{E_k} |f| \le \|f\|_{L^1} < \infty,$$

so $\mu(E_k) < \infty$ for every k. (If no a_j is non-zero, then $\varphi = 0$, and $\psi = g := 0$ work in Steps 2–4.)

Step 2: finite unions of open intervals. Recall the Proposition (Approximation by finite unions of intervals) in Part I, Section 9: for $E \in \mathcal{M}_\mu$ with $\mu(E) < \infty$ and any $\delta > 0$, there is a finite union A of open intervals with $\mu(E \triangle A) < \delta$. Since $E \triangle A = (E \setminus A) \cup (A \setminus E)$, we have $|\chi_E - \chi_A| = \chi_{E \triangle A}$, so

$$\mu(E \triangle A) = \int |\chi_E - \chi_A|.$$

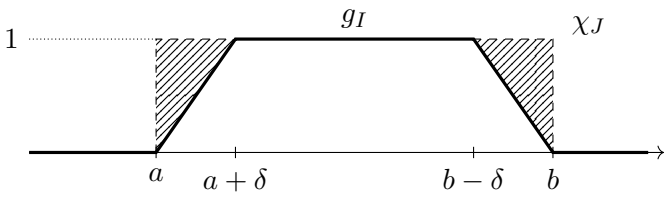
By Step 1 we can apply this to each E_j with $\delta := \varepsilon/(3\sum_j |a_j|)$. This gives finite unions A_j of open intervals, and we set $\psi := \sum_{j=1}^N a_j \chi_{A_j}$. Since $|\varphi - \psi| \le \sum_j |a_j||\chi_{E_j} - \chi_{A_j}|$,

$$\|\varphi - \psi\|_{L^1} \le \sum_{j=1}^N |a_j|\,\mu(E_j \triangle A_j) < \frac{\varepsilon}{3},$$

and so $\|f - \psi\|_{L^1} \le \|f - \varphi\|_{L^1} + \|\varphi - \psi\|_{L^1} < \frac{2\varepsilon}{3}$. This proves the first part of ⓑ.

Step 3: approximating one interval. Let I be an open interval with $\mu(I) < \infty$, and let $\eta > 0$. We construct $g_I \in C_c^0(\mathbb{R})$ with $0 \le g_I \le 1$ and $\|\chi_I - g_I\|_{L^1} < \eta$.

If I is unbounded, then $I \cap (-M, M) \nearrow I$ as $M \to \infty$, so by continuity from below and $\mu(I) < \infty$, $\mu\big(I \setminus (-M, M)\big) \to 0$. Choose M so large that $I \cap (-M, M) \neq \emptyset$ and $\mu\big(I \setminus (-M, M)\big) < \eta/2$, and let $J := I \cap (-M, M) = (a, b)$, a bounded open interval (if I is bounded, take $J := I$). For $0 < \delta < (b - a)/2$, let g_I be 0 outside (a, b), 1 on $[a + \delta, b - \delta]$, and linear on $[a, a + \delta]$ and on $[b - \delta, b]$:



Then g_I is continuous and vanishes outside the bounded interval (a,b). Moreover $0 \le \chi_J - g_I \le 1$, and $\chi_J - g_I = 0$ outside $(a, a+\delta) \cup (b-\delta, b)$ (hatched). Hence

$$\|\chi_J - g_I\|_{L^1} \le \mu\big((a,a+\delta)\big) + \mu\big((b-\delta,b)\big) \le \big[F(a+\delta) - F(a)\big] + \big[F(b^-) - F(b-\delta)\big],$$

using $\mu((a,a+\delta)) \le \mu((a,a+\delta]) = F(a+\delta) - F(a)$ and, by continuity from below, $\mu((b-\delta,b)) = \lim_k \mu((b-\delta, b-\frac{1}{k}]) = F(b^-) - F(b-\delta)$. As $\delta \searrow 0$, the first bracket tends to 0 because F is right continuous, and the second because $F(b-\delta) \to F(b^-)$. Choose δ with the right side $< \eta/2$. Since $|\chi_I - g_I| \le \chi_{I\setminus J} + |\chi_J - g_I|$,

$$\|\chi_I - g_I\|_{L^1} \le \mu(I \setminus J) + \|\chi_J - g_I\|_{L^1} < \eta.$$

Step 4: continuous approximation. A finite union of open intervals is a finite *disjoint* union of open intervals: two overlapping open intervals have an open interval as union, so overlapping ones can be merged until none overlap. Each interval $I \subseteq A_j$ obtained this way has $\mu(I) \le \mu(A_j) \le \mu(E_j) + \mu(E_j \triangle A_j) < \infty$. So $\psi = \sum_{l=1}^{L} b_l \chi_{I_l}$ for some open intervals I_l with $\mu(I_l) < \infty$ and scalars $b_l \neq 0$ (if $\psi = 0$, take $g := 0$). By Step 3 there are $g_l \in C_c^0(\mathbb{R})$ with $\|\chi_{I_l} - g_l\|_{L^1} < \varepsilon/(3\sum_l |b_l|)$. Then $g := \sum_{l=1}^{L} b_l g_l$ is continuous and vanishes outside a bounded interval (one containing the finitely many intervals outside which the g_l vanish), so $g \in C_c^0(\mathbb{R})$. Finally

$$\|\psi - g\|_{L^1} \le \sum_{l=1}^{L} |b_l| \, \|\chi_{I_l} - g_l\|_{L^1} < \frac{\varepsilon}{3},$$

so $\|f - g\|_{L^1} \le \|f - \psi\|_{L^1} + \|\psi - g\|_{L^1} < \varepsilon$.

Remark. In its sketch of Step 3, with $\delta = \varepsilon$ and bounded $I = (a,b)$, the lecture bounded $\mu_F([a,a+\varepsilon])$ and $\mu_F([b-\varepsilon,b])$ by $\sup_{[a,a+\varepsilon]} |F| \cdot \varepsilon$ and $\sup_{[b-\varepsilon,b]} |F| \cdot \varepsilon$. That would need a bound on the slope of F, and the closed intervals do not shrink to measure zero when F jumps at a: then $\mu_F([a,a+\varepsilon]) \ge F(a) - F(a^-) > 0$ for every ε. The proof above uses only the open intervals where $\chi_J \neq g_I$, and the right continuity of F.

Exercise (Smooth approximation)

The function g in (b) can even be taken in $C_c^\infty(\mathbb{R})$, i.e. smooth and 0 outside a bounded interval (the lecture remarks this). Prove it. (*Hint:* in Step 3, replace the linear pieces of g_I by smooth ones. With $h(x) := e^{-1/x}$ for $x > 0$ and $h(x) := 0$ for $x \le 0$, which is C^∞, the function $s(x) := h(x)/\big(h(x) + h(1-x)\big)$ is C^∞, takes values in $[0,1]$, and equals 0 for $x \le 0$ and 1 for $x \ge 1$.)

Next. Interchanging differentiation and integration; Lebesgue versus Riemann integration.

15 Applications of the Dominated Convergence Theorem

This lecture applies the DCT to integrals that depend on a parameter, and compares the Lebesgue integral with the Riemann integral.

15.1 Continuity and Differentiation under the Integral Sign

Theorem (Integrals depending on a parameter)

Let $(X, \mathcal{M}, \mu)$ be a measure space, $-\infty < a < b < \infty$, and $f : X \times [a,b] \to \mathbb{C}$ with $f(\cdot, t) \in L^1(\mu)$ for every $t \in [a,b]$. Set

$$F(t) := \int_X f(x,t)\, d\mu(x), \qquad t \in [a,b].$$

(a) Let $t_0 \in [a,b]$. Suppose there is $g \in L^1(\mu)$ with $|f(x,t)| \le g(x)$ for all $x \in X$ and $t \in [a,b]$, and that $\lim_{t\to t_0} f(x,t) = f(x,t_0)$ for every $x \in X$. Then $\lim_{t\to t_0} F(t) = F(t_0)$.

(b) Suppose $\frac{\partial f}{\partial t}(x,t)$ exists for all $x \in X$ and $t \in [a,b]$ (one-sided at a and b), and there is $g \in L^1(\mu)$ with

$$\left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x) \qquad \text{for all } x \in X,\ t \in [a,b].$$

Then F is differentiable on $[a,b]$ and

$$F'(t) = \int_X \frac{\partial f}{\partial t}(x,t)\, d\mu(x). \tag{$*$}$$

Proof (a)

It is enough to show that $F(t_n) \to F(t_0)$ for every sequence $t_n \in [a,b]$ with $t_n \to t_0$. Apply the DCT to $f_n(x) := f(x,t_n)$. By assumption $|f_n(x)| = |f(x,t_n)| \le g(x)$ with $g \in L^1$, and $f_n(x) \to f(x,t_0)$ for every x. So

$$\lim_{n\to\infty} F(t_n) = \lim_{n\to\infty} \int_X f(x,t_n)\, d\mu(x) = \int_X \lim_{n\to\infty} f(x,t_n)\, d\mu(x) = \int_X f(x,t_0)\, d\mu(x) = F(t_0).$$

Proof (b)

Fix $t_0 \in [a,b]$, and let $\{t_n\}_{n=1}^{\infty} \subseteq [a,b]$ with $t_n \to t_0$ and $t_n \neq t_0$. Set

$$h_n(x) := \frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}.$$

Each h_n is in L^1, by linearity (Section 14). By assumption $\lim_n h_n(x)$ exists for every x, and

$$\frac{\partial f}{\partial t}(x,t_0) = \lim_{n\to\infty} h_n(x),$$

so $\frac{\partial f}{\partial t}(\cdot, t_0) \in \mathrm{Meas}(X)$, as a pointwise limit of measurable functions (Part I, Section 10). By the mean value theorem,

$$|h_n(x)| = \left|\frac{f(x,t_n) - f(x,t_0)}{t_n - t_0}\right| \le \sup_{t\in[a,b]} \left|\frac{\partial f}{\partial t}(x,t)\right| \le g(x).$$

(For complex f, let $c := f(x,t_n) - f(x,t_0)$; if $c = 0$ there is nothing to prove. Otherwise let $u := \overline{\operatorname{sgn} c}$, so $|u| = 1$ and $|c| = \operatorname{Re}(uc)$. Then apply the mean value theorem to the real function $t \mapsto \operatorname{Re}\big(uf(x,t)\big)$, whose derivative $\operatorname{Re}\big(u\frac{\partial f}{\partial t}(x,t)\big)$ is bounded in absolute value by $g(x)$.)

So the DCT applies to $\{h_n\}$, with dominating function g. Using linearity,

$$\frac{F(t_n) - F(t_0)}{t_n - t_0} = \int_X h_n(x)\,d\mu(x) \longrightarrow \int_X \lim_{n\to\infty} h_n(x)\,d\mu(x) = \int_X \frac{\partial f}{\partial t}(x,t_0)\,d\mu(x).$$

This holds for every such sequence $t_n \to t_0$, so $F'(t_0)$ exists and is given by $(*)$.

Remark. In particular, if $f(x,\cdot)$ is continuous on $[a,b]$ for every x and $|f| \le g \in L^1$, then ⓐ at every t_0 shows that F is continuous on $[a,b]$. The lecture states the two parts this way: limits and integrals can be interchanged for f continuous in t, and differentiation and integration for f continuously differentiable in t, given a dominating g independent of t.

Explanation. The dominating function cannot be dropped in ⓐ. On $X = (0,\infty)$ with Lebesgue measure, let $f(x,t) := te^{-tx}$ for $t \in [0,1]$. Each $f(x,\cdot)$ is continuous and each $f(\cdot,t)$ is integrable, but $F(t) = \int_0^\infty te^{-tx}\,dx = 1$ for $t > 0$ while $F(0) = 0$. No integrable g dominates: for $x \ge 1$, $\sup_t f(x,t) \ge f(x,\frac{1}{x}) = \frac{1}{ex}$, which is not integrable on $[1,\infty)$.

15.2 Lebesgue versus Riemann Integration

Let $-\infty < a < b < \infty$ and let $f : [a,b] \to \mathbb{R}$ be bounded. Partitions P of $[a,b]$, the upper and lower sums $U(f,P)$ and $L(f,P)$, the upper and lower integrals $U(f)$ and $L(f)$, the class $\mathrm{RI}([a,b])$ and the Riemann integral $\int_a^b f(x)\,dx$ are as in Part I, Section 1. (The lecture writes $U_P f$, $L_P f$, U_f, L_f.) On $[a,b]$, m denotes Lebesgue measure on the Lebesgue measurable subsets of $[a,b]$.

Theorem (Lebesgue versus Riemann)

Let $f : [a,b] \to \mathbb{R}$ be bounded.

ⓐ If $f \in \mathrm{RI}([a,b])$, then f is Lebesgue measurable, $f \in L^1([a,b],m)$, and

$$\int_a^b f(x)\,dx = \int_{[a,b]} f\,dm.$$

ⓑ $f \in \mathrm{RI}([a,b])$ if and only if

$$m\big(\{x \in [a,b] \mid f \text{ is discontinuous at } x\}\big) = 0.$$

Proof (a)

Step functions. For a partition $P = \{a = t_0 < t_1 < \cdots < t_n = b\}$, let $I_1 := [t_0,t_1]$ and $I_j := (t_{j-1},t_j]$ for $2 \le j \le n$; these intervals are disjoint with union $[a,b]$. With $M_j := \sup_{[t_{j-1},t_j]} f$ and $m_j := \inf_{[t_{j-1},t_j]} f$, consider the simple functions

$$G_P := \sum_{j=1}^n M_j \chi_{I_j}, \qquad g_P := \sum_{j=1}^n m_j \chi_{I_j}.$$

Then $g_P \le f \le G_P$ on $[a,b]$, and since $m(I_j) = t_j - t_{j-1}$,

$$U(f,P) = \int_{[a,b]} G_P \, dm, \qquad L(f,P) = \int_{[a,b]} g_P \, dm.$$

If $P' \supseteq P$ is a refinement, then $G_{P'} \le G_P$ and $g_{P'} \ge g_P$. Indeed, each interval I'_i of P' lies in some I_j of P, and then $[t'_{i-1}, t'_i] \subseteq [t_{j-1}, t_j]$, so its supremum is at most M_j and its infimum at least m_j.

A nested sequence of partitions. Since $f \in \mathrm{RI}([a,b])$, $U(f) = L(f) = \int_a^b f(x)\,dx$. By the definitions of $U(f)$ and $L(f)$ there are partitions Q_k, R_k with $U(f,Q_k) < U(f) + \frac{1}{k}$ and $L(f,R_k) > L(f) - \frac{1}{k}$. Set $P_k := \bigcup_{i=1}^k (Q_i \cup R_i)$, so that $P_k \subseteq P_{k+1}$. Refining raises lower sums and lowers upper sums († in Part I, Section 1), so

$$L(f) - \tfrac{1}{k} < L(f,P_k) \le L(f) = U(f) \le U(f,P_k) < U(f) + \tfrac{1}{k}.$$

Hence $\lim_k U(f,P_k) = \lim_k L(f,P_k) = \int_a^b f(x)\,dx$.

The limits G and g. By the refinement property, G_{P_k} decreases and g_{P_k} increases as $k \to \infty$. Both are bounded by $M := \sup|f|$, so the limits

$$G := \lim_k G_{P_k}, \qquad g := \lim_k g_{P_k}$$

exist. They are measurable (Part I, Section 10), and

$$g \le f \le G. \tag{1}$$

The constant M lies in $L^1([a,b],m)$, since $m([a,b]) < \infty$, so the DCT applies to G_{P_k} and to g_{P_k}:

$$\int_{[a,b]} G \, dm = \lim_k U(f,P_k) = \int_a^b f(x)\,dx = \lim_k L(f,P_k) = \int_{[a,b]} g \, dm.$$

So $\int_{[a,b]} (G-g)\,dm = 0$ with $G - g \ge 0$, and $G = g$ a.e. by the Proposition (Zero integral) in Section 13. By (1), $f = g$ a.e. Since Lebesgue measure is complete, f is Lebesgue measurable (Part I, Section 11), and $f \in L^1$ as $|f| \le M$. Finally, by the Corollary (Null sets, integrable case) in Section 13,

$$\int_{[a,b]} f \, dm = \int_{[a,b]} g \, dm = \int_a^b f(x)\,dx.$$

Remark. The lecture defines G_P and g_P with the closed intervals $[t_{j-1}, t_j]$. These overlap at the partition points, where $g_P \le f \le G_P$ can fail: for $f \equiv 1$, $g_P = 2$ at every interior partition point. The integrals are unaffected. The lecture also writes $\int g_{P_k}\,dm \le \int_{[a,b]} f\,dm \le \int G_{P_k}\,dm$ before f is known to be measurable; the proof above avoids this. Finally, the lecture takes the monotonicity of G_{P_k} and g_{P_k} for granted. That needs nested partitions, which is why the P_k above are built as unions.

Exercise (Lebesgue's criterion)

Prove (b) of the Theorem (Lebesgue versus Riemann); the lecture left it as an exercise. (*Hint:* Let

$$H(x) := \lim_{\delta \searrow 0} \sup\{f(y) \mid y \in [a,b],\ |y-x| < \delta\},$$
$$h(x) := \lim_{\delta \searrow 0} \inf\{f(y) \mid y \in [a,b],\ |y-x| < \delta\}.$$

Unlike $\limsup_{y\to x} f(y)$ in Part I, Section 2, these allow $y = x$, so f is continuous at x iff $H(x) = h(x)$. The construction in the proof of (a) works for any bounded f, giving $U(f,P_k) \to U(f)$ and $L(f,P_k) \to L(f)$. Arrange in addition that $\max_j(t_j - t_{j-1}) \to 0$ along the P_k, and show that then $G = H$ and $g = h$ outside the countable set $\bigcup_k P_k$. Deduce that $U(f) - L(f) = \int_{[a,b]} (H-h)\,dm$.)

Explanation. Part (a) has no converse. $\chi_{\mathbb{Q}\cap[0,1]}$ (Example 2 in Part I, Section 2) is in L^1 with integral 0 but is not Riemann integrable, as (b) predicts: it is discontinuous everywhere. The Lebesgue integral also handles the unbounded $f(x) = 1/\sqrt{x}$ of Example 1 there. On $[\frac{1}{n}, 1]$, f is continuous, so by (a) and the MCT applied to $f\chi_{[1/n,1]} \nearrow f$ on $(0,1]$,

$$\int_{(0,1]} f\,dm = \lim_{n\to\infty} \int_{1/n}^{1} \frac{dx}{\sqrt{x}} = \lim_{n\to\infty} \left(2 - \frac{2}{\sqrt{n}}\right) = 2,$$

with no improper integral needed.

16 Modes of Convergence

16.1 Pointwise and L^1 Convergence

Let $(X, \mathcal{M}, \mu)$ be a measure space and $\{f_n\}_{n=1}^{\infty} \subseteq \mathrm{Meas}(X)$. So far there are two notions of convergence: a.e. pointwise convergence, and *convergence in* L^1, which means $\|f_n - f\|_{L^1} \to 0$ for $f_n, f \in L^1$ (Section 14).

Question. How are these related?

Counterexample (a.e. convergence without L^1 convergence)

On $(\mathbb{R}, \mathcal{L}, m)$, let $f_n := \frac{1}{n}\chi_{(0,n)}$. Then $f_n \to 0$ a.e. (in fact everywhere, and even uniformly, since $\sup|f_n| = \frac{1}{n}$). But

$$\int |f_n - 0|\,dm = \int |f_n|\,dm = \frac{1}{n}\,m\big((0,n)\big) = 1 \qquad \text{for all } n,$$

so $f_n \not\to 0$ in L^1.

Remark. The converse is also false in general: convergence in L^1 does not imply a.e. convergence.

Explanation. The standard example is the "typewriter" sequence on $[0,1]$: for $n = 2^k + j$ with $0 \le j < 2^k$, let $f_n := \chi_{[j2^{-k},(j+1)2^{-k}]}$. The intervals sweep across $[0,1]$ again and again, getting shorter each time. So $\|f_n\|_{L^1} = 2^{-k} \to 0$, yet every $x \in [0,1]$ lies in infinitely many of the intervals and outside infinitely many, so $f_n(x)$ converges for no x. The subsequence $f_{2^k} = \chi_{[0,2^{-k}]}$ does converge to 0 for every $x \neq 0$, as the Corollary (Subsequence) below predicts.

There are two positive results. The first is a consequence of the DCT.

Corollary (L^1 convergence from domination)

Let $f_n \to f$ a.e., and suppose $|f_n| \le g$ a.e. for all n, for some $g \in L^1$. Then $\|f_n - f\|_{L^1} \to 0$.

Proof

$|f| \le g$ a.e., as a limit, so $|f_n - f| \le |f_n| + |f| \le 2g$ a.e., with $2g \in L^1$. Since $|f_n - f| \to 0$ a.e., the DCT gives $\int |f_n - f| \to 0$.

Second, if $f_n \to f$ in L^1, then some subsequence converges to f a.e. (the Corollary (Subsequence) below). This is a compactness statement. To prove it, two more notions of convergence are useful.

16.2 Convergence in Measure

Definition

Let $\{f_n\}_{n=1}^{\infty} \subseteq \mathrm{Meas}(X)$ and $f \in \mathrm{Meas}(X)$.

ⓐ $\{f_n\}$ is *Cauchy in measure* if for every $\varepsilon > 0$,

$$\mu\big(\{x \mid |f_n(x) - f_m(x)| \ge \varepsilon\}\big) \to 0 \qquad \text{as } m, n \to \infty.$$

ⓑ $\{f_n\}$ *converges in measure* to f if for every $\varepsilon > 0$,

$$\mu\big(\{x \mid |f_n(x) - f(x)| \ge \varepsilon\}\big) \to 0 \qquad \text{as } n \to \infty.$$

Explanation. Convergence in measure allows f_n to be far from f, but only on a set whose measure tends to 0. The typewriter sequence above converges to 0 in measure, since $\mu(\{f_n \neq 0\}) = 2^{-k}$, but it converges nowhere. Convergence in measure says how much of X is bad for each n, not where the bad set is.

Proposition (L^1 implies measure)

Suppose $f_n \to f$ in L^1. Then $f_n \to f$ in measure.

Proof

Let $\varepsilon > 0$ and $E_{n,\varepsilon} := \{x \mid |f_n(x) - f(x)| \ge \varepsilon\}$. Since $|f_n - f| \ge \varepsilon$ on $E_{n,\varepsilon}$,

$$\|f_n - f\|_{L^1} = \int_X |f_n - f| \ge \int_{E_{n,\varepsilon}} |f_n - f| \ge \varepsilon\, \mu(E_{n,\varepsilon}).$$

So $\mu(E_{n,\varepsilon}) \le \frac{1}{\varepsilon}\|f_n - f\|_{L^1} \to 0$ as $n \to \infty$. (This is the argument of Chebyshev's inequality,

Section 13.)

Theorem (Cauchy in measure)

Suppose $\{f_n\}_{n=1}^{\infty} \subseteq \mathrm{Meas}(X)$ is Cauchy in measure. Then there is $f \in \mathrm{Meas}(X)$ such that

(i) $f_n \to f$ in measure;
(ii) there is a subsequence $\{f_{n_j}\}_{j=1}^{\infty}$ of $\{f_n\}$ with $f_{n_j} \to f$ a.e.

Moreover, if also $f_n \to g$ in measure, then $g = f$ a.e.

Proof

Choosing the subsequence. Since $\{f_n\}$ is Cauchy in measure, for each j there is N_j with

$$\mu\big(\{x \mid |f_n(x) - f_m(x)| \geq 2^{-j}\}\big) \leq 2^{-j} \qquad \text{for all } m, n \geq N_j.$$

Choose $n_1 < n_2 < \cdots$ with $n_j \geq N_j$, and set $g_j := f_{n_j}$. Since $n_{j+1} > n_j \geq N_j$, the sets

$$E_j := \{x \mid |g_j(x) - g_{j+1}(x)| \geq 2^{-j}\}$$

satisfy $\mu(E_j) \leq 2^{-j}$. Set $F_k := \bigcup_{j=k}^{\infty} E_j$. Then

$$\mu(F_k) \leq \sum_{j=k}^{\infty} 2^{-j} = 2^{1-k},$$

so $\mu(F_k) \to 0$ as $k \to \infty$.

Behaviour of g_j off F_k. Let $x \notin F_k$. Then $x \notin E_l$ for every $l \geq k$, i.e. $|g_{l+1}(x) - g_l(x)| < 2^{-l}$. So for $i > j \geq k$,

$$|g_j(x) - g_i(x)| \leq \sum_{l=j}^{i-1} |g_{l+1}(x) - g_l(x)| \leq \sum_{l=j}^{i-1} 2^{-l} \leq 2^{1-j}. \tag{$*$}$$

Hence $\{g_j(x)\}_j$ is a Cauchy sequence for every $x \notin F_k$. Let

$$F := \bigcap_{k=1}^{\infty} F_k = \limsup_j E_j.$$

The F_k decrease and $\mu(F_1) \leq 1 < \infty$, so by continuity from above $\mu(F) = \lim_k \mu(F_k) = 0$.

The limit. Every $x \notin F$ lies outside some F_k, so $\lim_j g_j(x)$ exists there. Set

$$f(x) := \begin{cases} \lim_j g_j(x), & x \notin F, \\ 0, & x \in F, \end{cases}$$

i.e. $f = \lim_j (g_j \chi_{F^c})$ pointwise on all of X. So $f \in \mathrm{Meas}(X)$, as a pointwise limit of measurable functions (Part I, Section 10). Since $\mu(F) = 0$, $g_j \to f$ a.e., which proves (ii).

$g_j \to f$ in measure. Let $i \to \infty$ in $(*)$: for $x \notin F_k$ and $j \geq k$,

$$|g_j(x) - f(x)| \leq 2^{1-j}.$$

Given $\varepsilon > 0$, for every j with $2^{1-j} < \varepsilon$ apply this with $k = j$: then $\{|g_j - f| \geq \varepsilon\} \subseteq F_j$, so

$$\mu\big(\{|g_j - f| \geq \varepsilon\}\big) \leq \mu(F_j) \leq 2^{1-j} \to 0.$$

$f_n \to f$ *in measure.* Let $\varepsilon > 0$. By the triangle inequality, for all n and j,

$$\{x \mid |f_n(x) - f(x)| \geq \varepsilon\} \subseteq \{x \mid |f_n(x) - g_j(x)| \geq \tfrac{\varepsilon}{2}\} \cup \{x \mid |g_j(x) - f(x)| \geq \tfrac{\varepsilon}{2}\}. \qquad (\dagger)$$

Given $\delta > 0$, choose N with $\mu(\{|f_n - f_m| \geq \frac{\varepsilon}{2}\}) < \delta$ for $m, n \geq N$, since $\{f_n\}$ is Cauchy in measure. Then choose j with $n_j \geq N$ and $\mu(\{|g_j - f| \geq \frac{\varepsilon}{2}\}) < \delta$, since $g_j \to f$ in measure. By $(\dagger)$, $\mu(\{|f_n - f| \geq \varepsilon\}) < 2\delta$ for all $n \geq N$. This proves (i).

Uniqueness. Suppose also $f_n \to g$ in measure, and let $\varepsilon > 0$. As in $(\dagger)$,

$$\{x \mid |f(x) - g(x)| \geq \varepsilon\} \subseteq \{x \mid |f(x) - f_n(x)| \geq \tfrac{\varepsilon}{2}\} \cup \{x \mid |f_n(x) - g(x)| \geq \tfrac{\varepsilon}{2}\}$$

for every n, and the measures of both sets on the right tend to 0 as $n \to \infty$. So $\mu(\{|f-g| \geq \varepsilon\}) = 0$ for every $\varepsilon > 0$. Taking $\varepsilon = \frac{1}{k}$ and the union over k, $\mu(\{f \neq g\}) = 0$.

Corollary (Subsequence)

Suppose $f_n \to f$ in $L^1(X, \mathcal{M}, \mu)$. Then there is a subsequence $\{f_{n_j}\}_{j=1}^{\infty}$ of $\{f_n\}$ with $f_{n_j} \to f$ a.e.

Proof

The lecture left this as an exercise. By the Proposition (L^1 implies measure), $f_n \to f$ in measure. Then $\{f_n\}$ is Cauchy in measure, since

$$\{|f_n - f_m| \geq \varepsilon\} \subseteq \{|f_n - f| \geq \tfrac{\varepsilon}{2}\} \cup \{|f - f_m| \geq \tfrac{\varepsilon}{2}\}.$$

By the Theorem (Cauchy in measure) there are $\tilde{f} \in \mathrm{Meas}(X)$ and a subsequence with $f_{n_j} \to \tilde{f}$ a.e. and $f_n \to \tilde{f}$ in measure. By uniqueness in that theorem, $\tilde{f} = f$ a.e., so $f_{n_j} \to f$ a.e.

16.3 Egoroff's Theorem

The lecture announces two more theorems worth discussing; only the first, Egoroff's theorem, is stated in this lecture.

Theorem (Egoroff)

Let $(X, \mathcal{M}, \mu)$ be a measure space with $\mu(X) < \infty$, and let $\{f_n\}_{n=1}^{\infty} \subseteq \mathrm{Meas}(X)$ with $f_n \to f$ a.e. Then for every $\varepsilon > 0$ there is $E_\varepsilon \in \mathcal{M}$ with $\mu(E_\varepsilon) < \varepsilon$ such that $f_n \to f$ uniformly on E_ε^c. We say that $f_n \to f$ *almost uniformly.*

Explanation. The hypothesis $\mu(X) < \infty$ cannot be dropped. On $(\mathbb{R}, \mathcal{L}, m)$, $f_n := \chi_{[n,n+1)} \to 0$ everywhere (Counterexample (Escaping mass) in Section 13). If $f_n \to 0$ uniformly on E^c, then $f_n < 1$ on E^c for all large n, i.e. $[n, n+1) \subseteq E$ for all large n, so $m(E) = \infty$. Also, "almost uniformly" does not mean "uniformly off a null set". On $[0,1]$, $x^n \to 0$ a.e., and uniformly on $[0, 1-\delta]$ for each $\delta > 0$. But it does not converge uniformly on $[0,1] \setminus N$ for any null set N, since

$[0,1) \setminus N$ has points arbitrarily close to 1, where x^n is close to 1.

Remark. Let (X,d) be a metric space with a measure μ, defined on a σ-algebra containing the Borel sets, with $\mu(X) < \infty$. Suppose $f_n \in C^0(X)$ for $n \geq 1$ and $f_n \to f$ a.e. By Egoroff's theorem, $f_n \to f$ almost uniformly. So for every $\varepsilon > 0$ there is E_ε with $\mu(E_\varepsilon) < \varepsilon$ such that f, restricted to E_ε^c, is a uniform limit of continuous functions, hence continuous. In this sense f is continuous off a set of arbitrarily small measure.

Next. The proof of Egoroff's theorem.